

To simulate the order book's evolution, we will follow the **Price/Time Priority** matching rules detailed in the provided resources. In this system, orders are prioritized first by the best price (highest for buys, lowest for sells) and secondarily by the time they were placed.

Based on the current **BTC/USDT** context from Binance and the system design principles discussed in the video, we will use a base price of **96,000 USDT**.

Simulation Parameters

- **Asset:** BTC/USDT
- **Matching Engine:** Price/Time Priority
- **Initial State:**
 - **Asks (Sells):** [S1: 1.0 BTC @ 96,100], [S2: 2.0 BTC @ 96,200]
 - **Bids (Buys):** [B1: 1.0 BTC @ 95,900], [B2: 2.0 BTC @ 95,800]

Order Simulation (15 Limit + 10 Market)

#	Order Type	Side	Qty	Price	Result / Action	Queue State (Top Levels)
1	Limit	Buy	0.5	95,950	Queued (New top bid)	Bids: [0.5@95,950], [1@95,900] Asks: [1@96,100], [2@96,200]
2	Limit	Buy	1.2	96,100	MATCH: 1.0 fills S1 @ 96,100. 0.2 remains at 96,100.	Bids: [0.2@96,100], [0.5@95,950] Asks: [2@96,200]
3	Market	Sell	0.8	—	MATCH: Fills 0.2 (Bid@96,100) then 0.5 (Bid@95,950) then 0.1 (Bid@95,900).	Bids: [0.9@95,900], [2@95,800] Asks: [2@96,200]

4	Limit	Sell	0.3	96,050	Queued (New top ask)	Bids: [0.9@95,900], [2@95,800] Asks: [0.3@96,050], [2@96,200]
5	Limit	Sell	1.0	96,100	Queued (Level 2 ask)	Bids: [0.9@95,900] Asks: [0.3@96,050], [1@96,100]
6	Market	Buy	0.4	—	MATCH: Fills 0.3 (Ask@96,050) then 0.1 (Ask@96,100).	Bids: [0.9@95,900] Asks: [0.9@96,100], [2@96,200]
7	Limit	Buy	0.8	95,950	Queued (New top bid)	Bids: [0.8@95,950], [0.9@95,900] Asks: [0.9@96,100]
8	Market	Sell	1.0	—	MATCH: Fills 0.8 (Bid@95,950) then 0.2 (Bid@95,900).	Bids: [0.7@95,900], [2@95,800] Asks: [0.9@96,100]
9	Limit	Sell	0.2	96,020	Queued (New top ask)	Bids: [0.7@95,900] Asks: [0.2@96,020], [0.9@96,100]

10	Market	Buy	0.5	—	MATCH: Fills 0.2 (Ask@96,020) then 0.3 (Ask@96,100).	Bids: [0.7@95,900] Asks: [0.6@96,100], [2@96,200]
11	Limit	Buy	1.5	96,000	Queued (New top bid)	Bids: [1.5@96,000], [0.7@95,900] Asks: [0.6@96,100]
12	Limit	Sell	1.0	96,030	Queued (New top ask)	Bids: [1.5@96,000] Asks: [1@96,030], [0.6@96,100]
13	Market	Sell	0.6	—	MATCH: Fills 0.6 of Bid@96,000.	Bids: [0.9@96,000], [0.7@95,900] Asks: [1@96,030]
14	Limit	Buy	0.2	95,980	Queued (Between top bids)	Bids: [0.9@96,000], [0.2@95,980], [0.7@95,900]
15	Market	Buy	1.2	—	MATCH: Fills 1.0 (Ask@96,030) then 0.2 (Ask@96,100).	Bids: [0.9@96,000] Asks: [0.4@96,100], [2@96,200]
16	Limit	Sell	0.5	96,150	Queued (Level 2 ask)	Bids: [0.9@96,000]

						Asks: [0.4@96,100], [0.5@96,150], [2@96,200]
17	Limit	Buy	1.0	95,950	Queued (Level 3 bid)	Bids: [0.9@96,000], [0.2@95,980], [1@95,950]
18	Market	Sell	2.0	—	MATCH: Fills 0.9 (96k), 0.2 (95.98k), 0.7 (95.9k), then 0.2 of 95.95k.	Bids: [0.8@95,950], [2@95,800] Asks: [0.4@96,100]
19	Limit	Sell	0.4	96,100	Queued (Time priority: joins end of 96,100 list)	Asks: [0.4@96,100 (Older)], [0.4@96,100 (Newer)]
20	Market	Buy	0.8	—	MATCH: Fills both asks at 96,100.	Bids: [0.8@95,950] Asks: [0.5@96,150], [2@96,200]
21	Limit	Buy	0.3	96,050	Queued (New top bid)	Bids: [0.3@96,050], [0.8@95,950] Asks: [0.5@96,150]
22	Market	Sell	0.5	—	MATCH: Fills 0.3 (96,050) then 0.2 (95,950).	Bids: [0.6@95,950], [2@95,800]

						Asks: [0.5@96,150]
23	Limit	Sell	0.7	96,120	Queued (New top ask)	Bids: [0.6@95,950] Asks: [0.7@96,120], [0.5@96,150]
24	Market	Buy	1.0	—	MATCH: Fills 0.7 (96,120) then 0.3 (96,150).	Bids: [0.6@95,950] Asks: [0.2@96,150], [2@96,200]
25	Market	Sell	1.5	—	MATCH: Fills 0.6 (95,950) then 0.9 (95,800).	Bids: [1.1@95,800] Asks: [0.2@96,150]

Summary of Final Queue State

After 15 limit orders and 10 market orders, the final state of the top of the order book is:

- **Best Bid:** 1.1 BTC @ 95,800 USDT
- **Best Ask:** 0.2 BTC @ 96,150 USDT

Key Observations:

- **Price Discovery:** Market orders consumed liquidity at various price points, causing the spread to shift.
- **Time Priority:** In Step 19, the new sell order at 96,100 was placed behind the existing sell order at that same price, as per standard matching protocols.
- **Market Impact:** Large market orders (like Step 18) "walked the book," filling multiple price levels and increasing execution costs (market impact).